Sampling variability

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

One complete population, sample and statistic record.

Keep IDs, selection rule, complete samples, means, weights and limits on transferred inference together.

Keep one habit: Recall → W1–W3/W5 → independent new-value case → sampling-variability project → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Confounders and controls. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Baselines and fair comparisons. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Replication and reproduction. Give an example and a mistake that the idea helps you avoid.

Confounders and controls

My remembered idea, example and correction

Baselines and fair comparisons

My remembered idea, example and correction

Replication and reproduction

My remembered idea, example and correction

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Sampling variability means different eligible samples can produce different statistics under fixed population/procedure conditions. Complete known model: U1 and U2 each have value 1; U3 and U4 each have value 3. Choose two distinct IDs uniformly without replacement. Six unordered pairs are equally likely: U1/U2 has mean 1; U1/U3, U1/U4, U2/U3 and U2/U4 have mean 2; U3/U4 has mean 3.

  The average of all six means is 2, equal to the known population mean. Four of six pairs have mean 2, so its probability is 4/6 = 2/3. Equal values retain distinct IDs. Enumeration lists possibilities, not six independent observed experiments; draws within a sample without replacement are not independent.

  Separate picture fixed classifier/rubric scores 3/4 = 75% and 2/4 = 50% on different four-case samples, 25 percentage points apart. No sampling model or confidence interval is supplied for that picture. Identical deterministic records/predictions/scoring should replay consistently. Code uses exact arithmetic on supplied fixtures, no random draw, live classifier or broad population inference.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

W2 · Trace the worked case

> Keep the supplied inputs, comparison rule and evidence together.

Complete task and evidence

- Sampling variability means different eligible samples can produce different statistics under fixed population/procedure conditions. Complete known model: U1 and U2 each have value 1; U3 and U4 each have value 3. Choose two distinct IDs uniformly without replacement. Six unordered pairs are equally likely: U1/U2 has mean 1; U1/U3, U1/U4, U2/U3 and U2/U4 have mean 2; U3/U4 has mean 3.

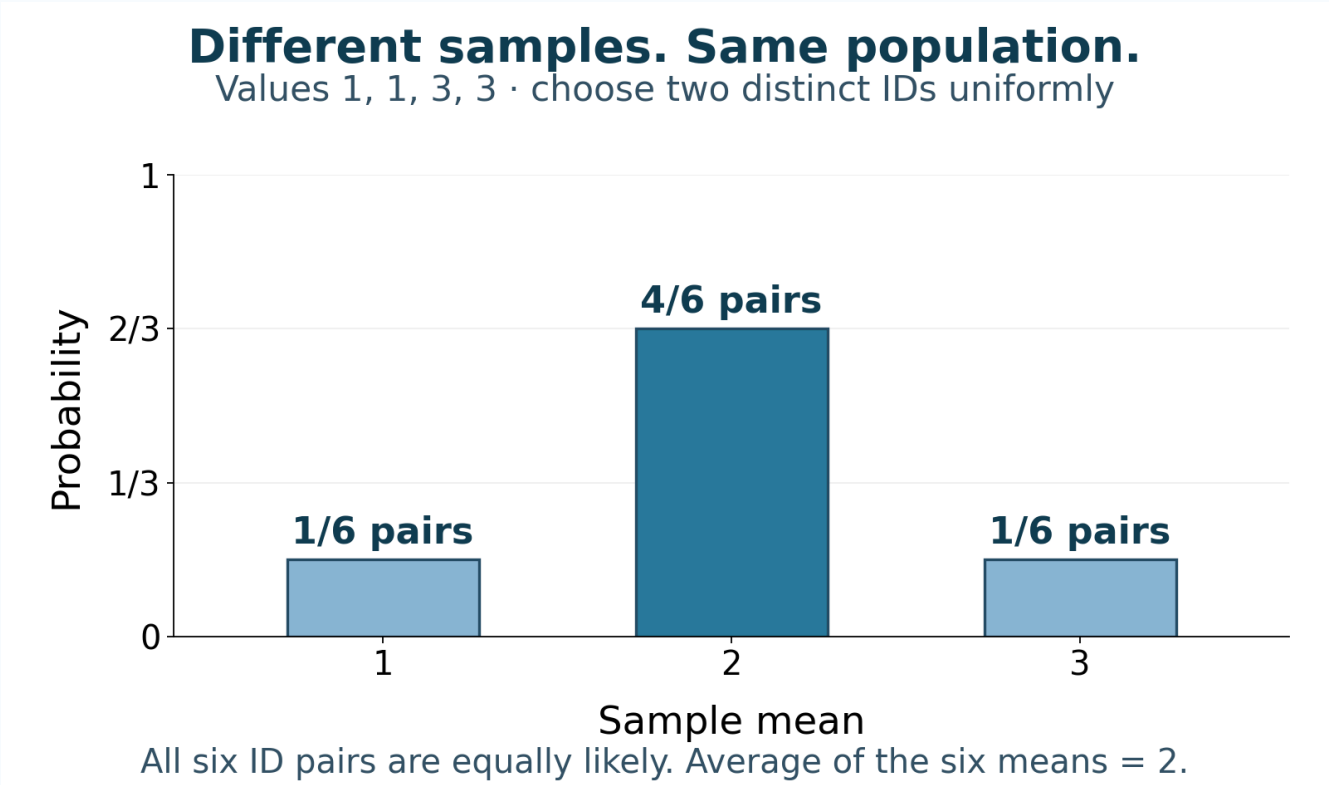
  The average of all six means is 2, equal to the known population mean. Four of six pairs have mean 2, so its probability is 4/6 = 2/3. Equal values retain distinct IDs. Enumeration lists possibilities, not six independent observed experiments; draws within a sample without replacement are not independent.

  Separate picture fixed classifier/rubric scores 3/4 = 75% and 2/4 = 50% on different four-case samples, 25 percentage points apart. No sampling model or confidence interval is supplied for that picture. Identical deterministic records/predictions/scoring should replay consistently. Code uses exact arithmetic on supplied fixtures, no random draw, live classifier or broad population inference.

- Population values are 1, 1, 3, 3, with four distinct IDs. Choose two IDs uniformly without replacement. List all six unordered pairs and their means, then find the mean of those means and the chance of obtaining a mean of 2. Show the intermediate steps and give the conclusion with its scope.

Unordered ID pair	Selected values	Sample mean	Model probability
U1, U2	1, 1	1	1/6
U1, U3	1, 3	2	1/6
U1, U4	1, 3	2	1/6
U2, U3	1, 3	2	1/6

Unordered ID pair	Selected values	Sample mean	Model probability
U2, U4	1, 3	2	1/6
U3, U4	3, 3	3	1/6
Summary	Six equally likely ID pairs	Average = (1+2+2+2+2+3)/6 = 2	P(mean 2) = 4/6 = 2/3



My reasoning and evidence

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Sampling variability means different eligible samples can produce different statistics under fixed population/procedure conditions. Complete known model: U1 and U2 each have value 1; U3 and U4 each have value 3. Choose two distinct IDs uniformly without replacement. Six unordered pairs are equally likely: U1/U2 has mean 1; U1/U3, U1/U4, U2/U3 and U2/U4 have mean 2; U3/U4 has mean 3.

  The average of all six means is 2, equal to the known population mean. Four of six pairs have mean 2, so its probability is 4/6 = 2/3. Equal values retain distinct IDs. Enumeration lists possibilities, not six independent observed experiments; draws within a sample without replacement are not independent.

  Separate picture fixed classifier/rubric scores 3/4 = 75% and 2/4 = 50% on different four-case samples, 25 percentage points apart. No sampling model or confidence interval is supplied for that picture. Identical deterministic records/predictions/scoring should replay consistently. Code uses exact arithmetic on supplied fixtures, no random draw, live classifier or broad population inference.

- A report omits this qualification: This finite example is not a general normality guarantee or a model of a biased convenience sample.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Sampling variability means different eligible samples can produce different statistics under fixed population/procedure conditions. Complete known model: U1 and U2 each have value 1; U3 and U4 each have value 3. Choose two distinct IDs uniformly without replacement. Six unordered pairs are equally likely: U1/U2 has mean 1; U1/U3, U1/U4, U2/U3 and U2/U4 have mean 2; U3/U4 has mean 3.

  The average of all six means is 2, equal to the known population mean. Four of six pairs have mean 2, so its probability is 4/6 = 2/3. Equal values retain distinct IDs. Enumeration lists possibilities, not six independent observed experiments; draws within a sample without replacement are not independent.

  Separate picture fixed classifier/rubric scores 3/4 = 75% and 2/4 = 50% on different four-case samples, 25 percentage points apart. No sampling model or confidence interval is supplied for that picture. Identical deterministic records/predictions/scoring should replay consistently. Code uses exact arithmetic on supplied fixtures, no random draw, live classifier or broad population inference.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Population values are 0, 0, 4, 4, with four distinct IDs. Choose two IDs uniformly without replacement. Enumerate all six unordered pairs, calculate their sample means and find their average.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

My independent explanation, trace and limits

MAIN LESSON · SECTION 10

Project · Keep the research evidence

> Preserve the question, protocol, observations, analysis and limits.

Complete task and evidence

- Add a labelled **sampling variability evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **sampling variability**.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for **Sampling variability**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for **Sampling variability**. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Population values are 0, 0, 4, 4, with four distinct IDs. Choose two IDs uniformly without replacement. Enumerate all six unordered pairs, calculate their sample means and find their average.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Two small samples can differ. A fixed classifier is scored on two distinct four-case samples from a declared collection.

- Sample A has 3 correct of 4; sample B has 2 correct of 4. The model and rubric are unchanged.

- Calculate the two four-case rates and their difference. Explain how changed included cases can affect a fixed model, and why identical deterministic records should not differ on a rerun.

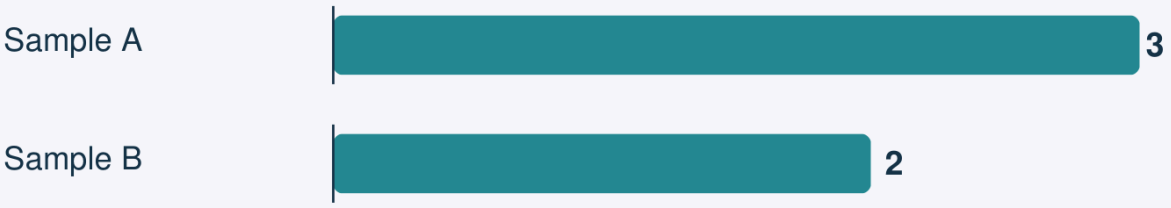
- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

  Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

AI ACADEMY / AI-7/11

Two distinct samples, fixed model

correct cases out of 4 in each sample



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/11

Sampling variability

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Keep model fixed

No training or threshold change occurs between samples.

↓

2. Compute observed rates

A is 75%; B is 50%, differing by 25 percentage points.

↓

3. Consider sampling

Different included cases can change a small-sample rate.

CHANGE ONE THING

Evaluate the exact same fixed four cases twice with deterministic predictions.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Reproducible AI Research Study
- Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.
- For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Two small samples can differ. A fixed classifier is scored on two distinct four-case samples from a declared collection.
- Which labelled step matches this detail? Different included cases can change a small-sample rate.
- 1. Consider sampling
- 2. Keep model fixed
- 3. Compute observed rates
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Consider sampling Different included cases can change a small-sample rate. **Sampling variability** means a statistic can vary because different units were observed.

Do not automatically attribute every rate change to a model improvement or deterioration.

Changed-case question and optional hint

- Changed-case practice: Evaluate the exact same fixed four cases twice with deterministic **prediction**s. Which conclusion is supported by the supplied facts?
- 1. The counts should agree unless some condition, data or scoring changed.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The counts should agree unless some condition, data or scoring changed. The sampling explanation no longer applies when the records are identical.

Repair and retain the record

Review Confounders and controls (ai-7/10). Goal: Explain the **mechanism** and terminology of confounders and controls. Short practice: Draw the four condition/background combinations.

Which variables were coupled before, and why would collecting only healthy-white and spotted-blue again leave the confounding unresolved?. Return to this week and explain what you changed.

Review Baselines and fair comparisons (ai-7/8). Goal: Explain the **mechanism** and terminology of baselines and fair comparisons. Short practice: Calculate 6/8 and 7/8 correctness and name the one-case gain.

Why is giving B the answer key incompatible with an unseen comparison to the simple baseline?. Return to this week and explain what you changed.

Review Replication and reproduction (ai-7/5). Goal: Explain the **mechanism** and terminology of replication and reproduction. Short practice: Calculate the scores from the original and new four-record sets.

Which activity reproduces saved analysis, which repeats the protocol, and why does a different rerun of the original data need investigation?. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

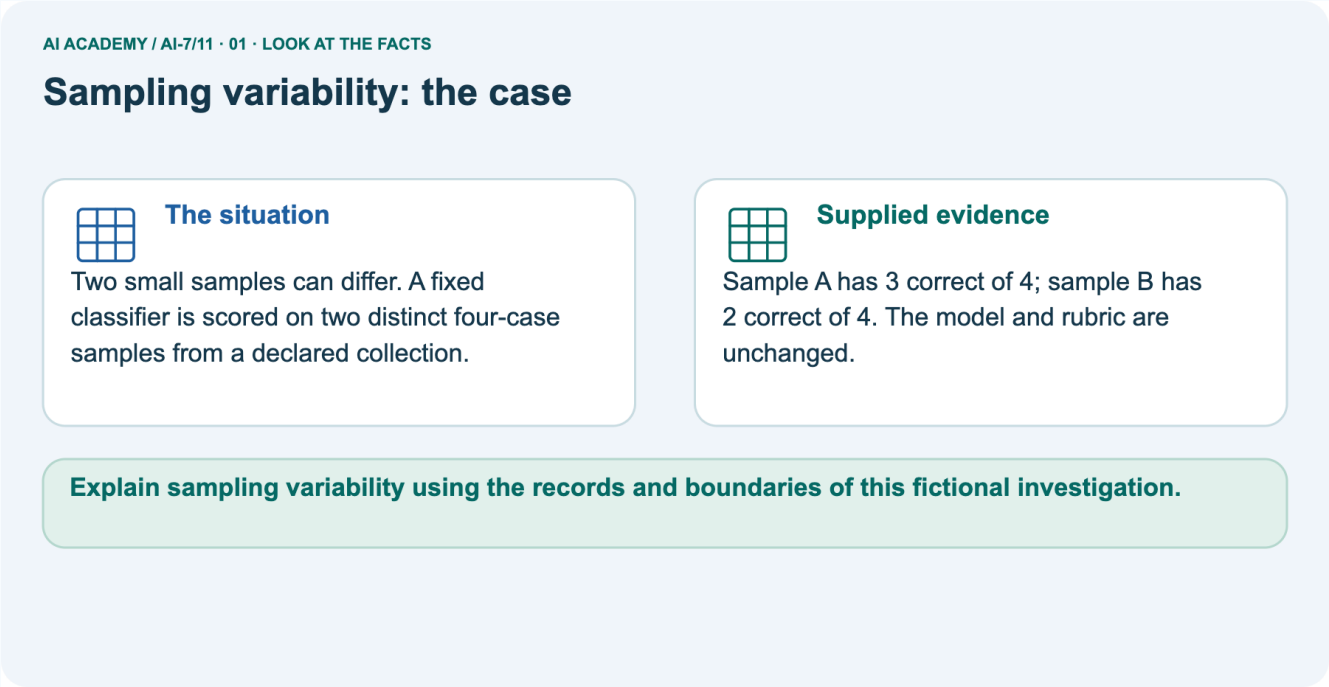
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Sampling variability: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

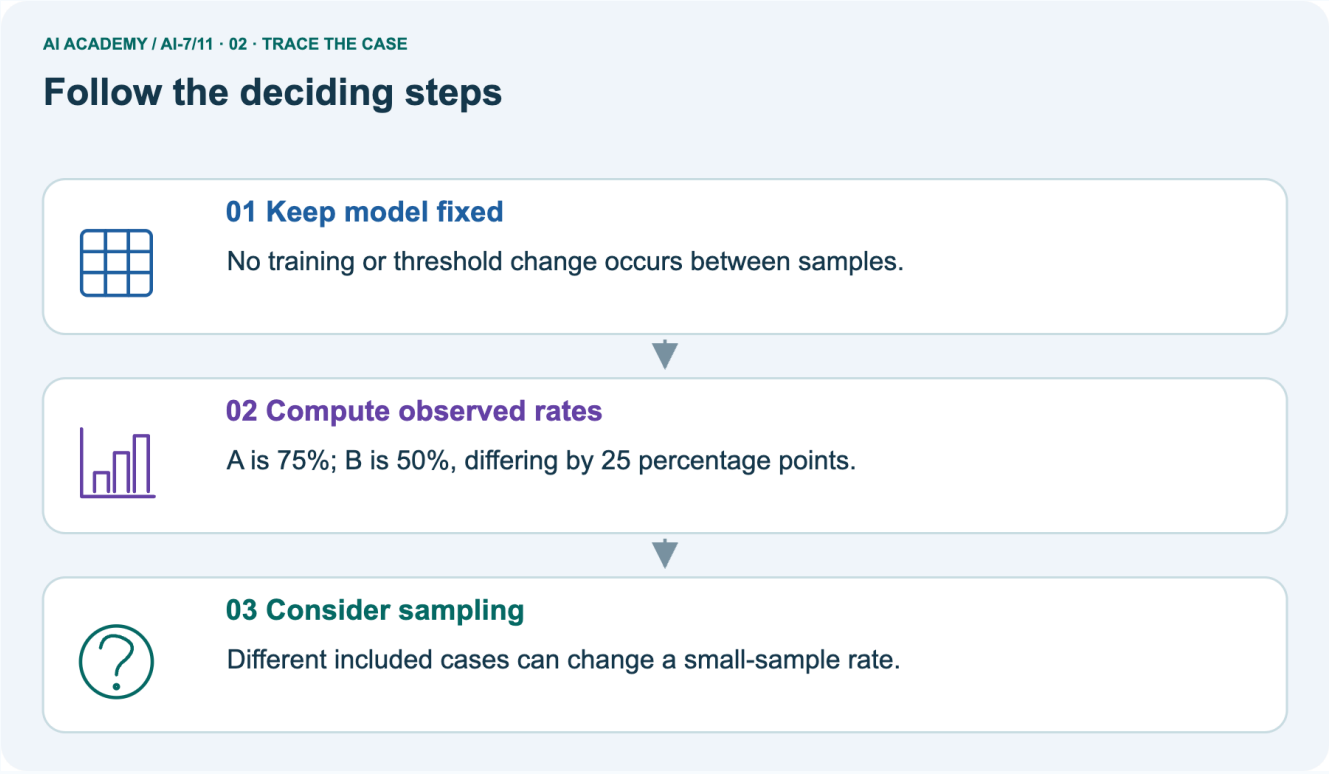
Two small samples can differ. A fixed classifier is scored on two distinct four-case samples from a declared collection. Supplied facts: Sample A has 3 correct of 4; sample B has 2 correct of 4.

The model and rubric are unchanged.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



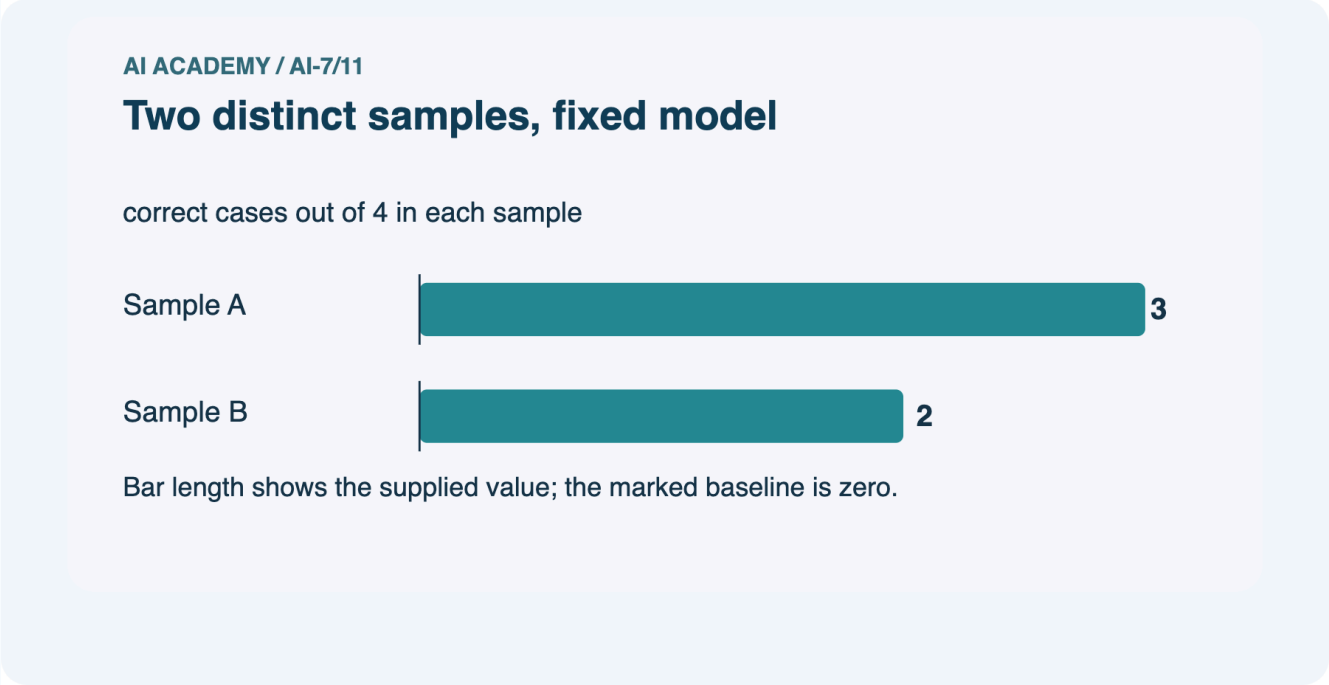
Follow the labelled picture with the matching explanation. Open the vector diagram

Keep model fixed: No **training** or threshold change occurs between samples. → Compute observed rates: A is 75%; B is 50%, differing by 25 percentage points. → Consider sampling: Different included cases can change a small-sample rate.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Two distinct samples, fixed model



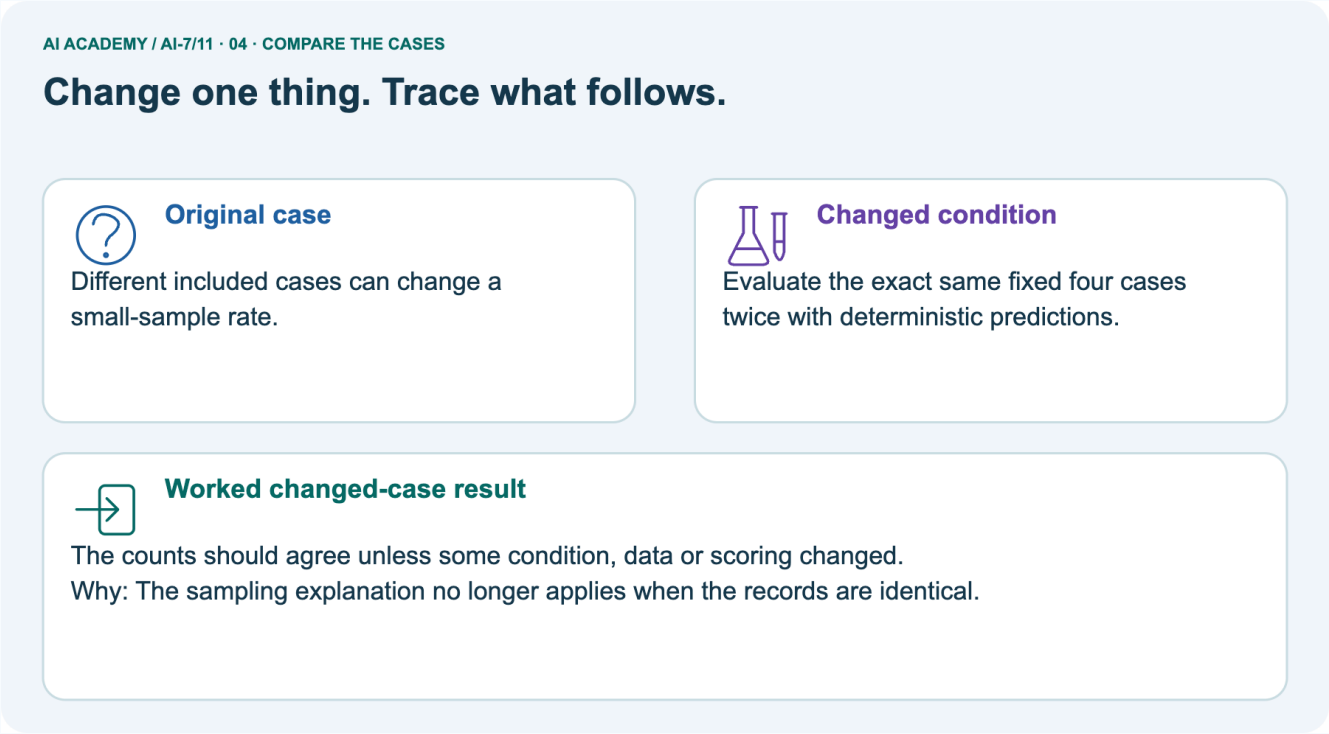
Follow the labelled picture with the matching explanation. Open the vector diagram

Two distinct samples, fixed model. Units: correct cases out of 4 in each sample. Sample A: 3. Sample B: 2.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

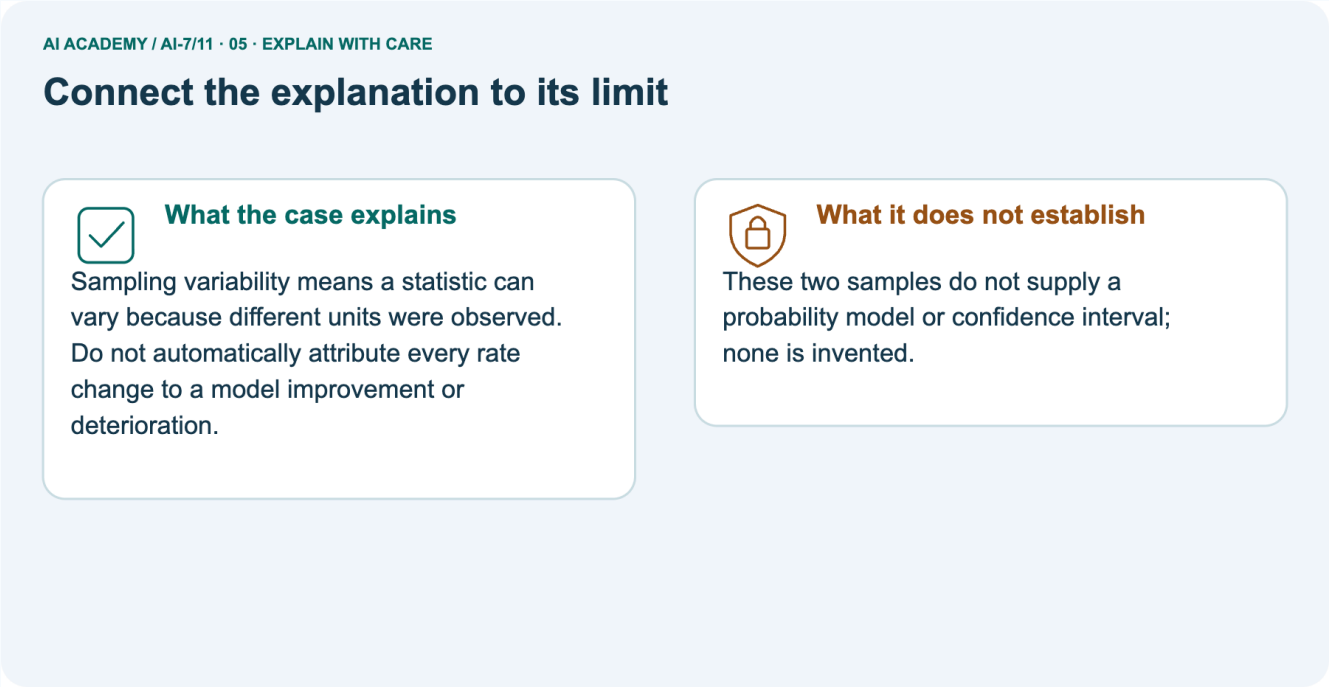
Original case: Different included cases can change a small-sample rate. Changed condition: Evaluate the exact same fixed four cases twice with deterministic **prediction**s.

Expected result: The counts should agree unless some condition, data or scoring changed. Reason: The sampling explanation no longer applies when the records are identical.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

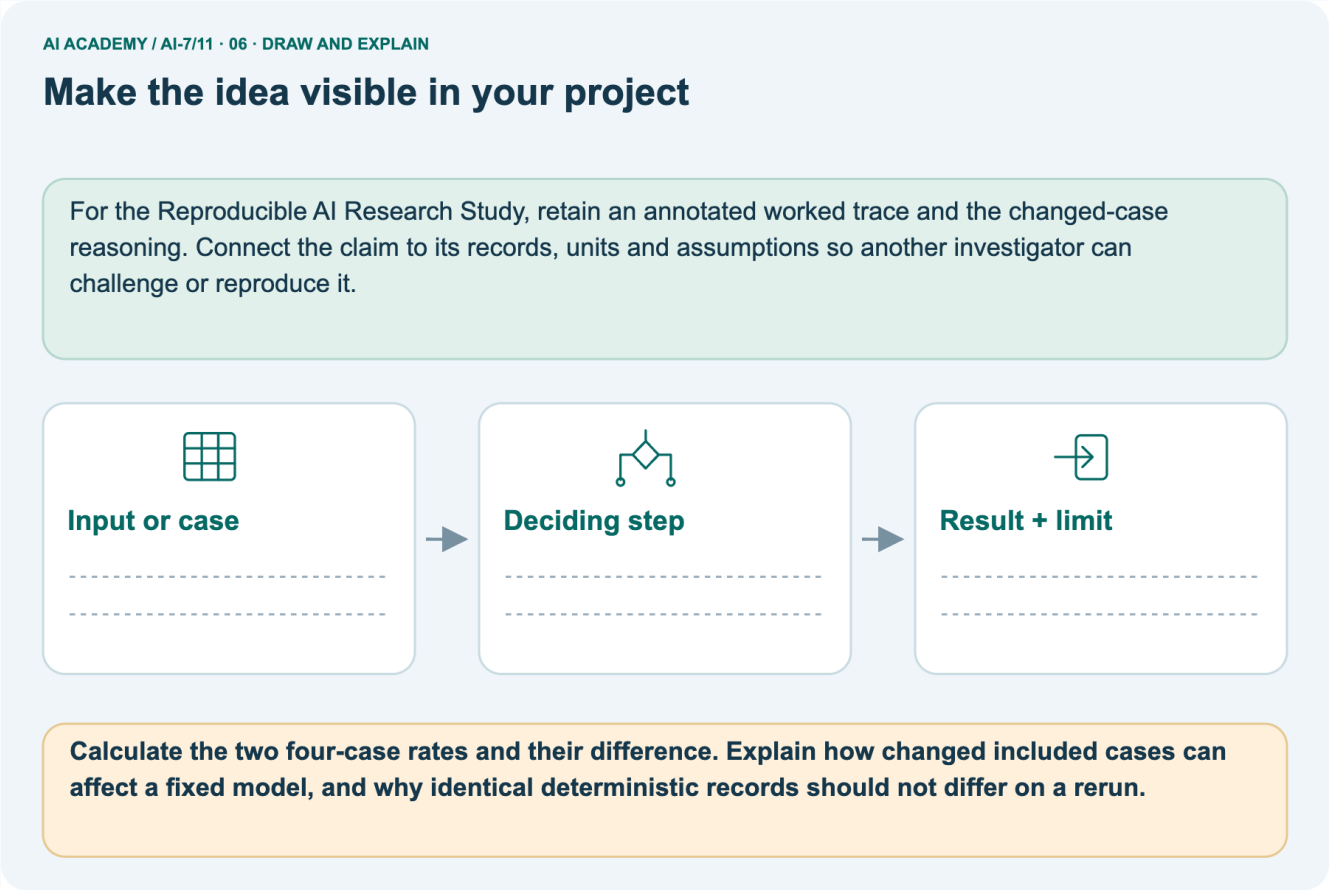
Sampling variability means a statistic can vary because different units were observed. Do not automatically attribute every rate change to a model improvement or deterioration.

Limit: These two samples do not supply a probability model or confidence interval; none is invented.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate the two four-case rates and their difference.

Explain how changed included cases can affect a fixed model, and why identical deterministic records should not differ on a rerun.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Sampling variability: the case

02. Follow the deciding steps

03. Two distinct samples, fixed model

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.